

MODEL EVALUATION

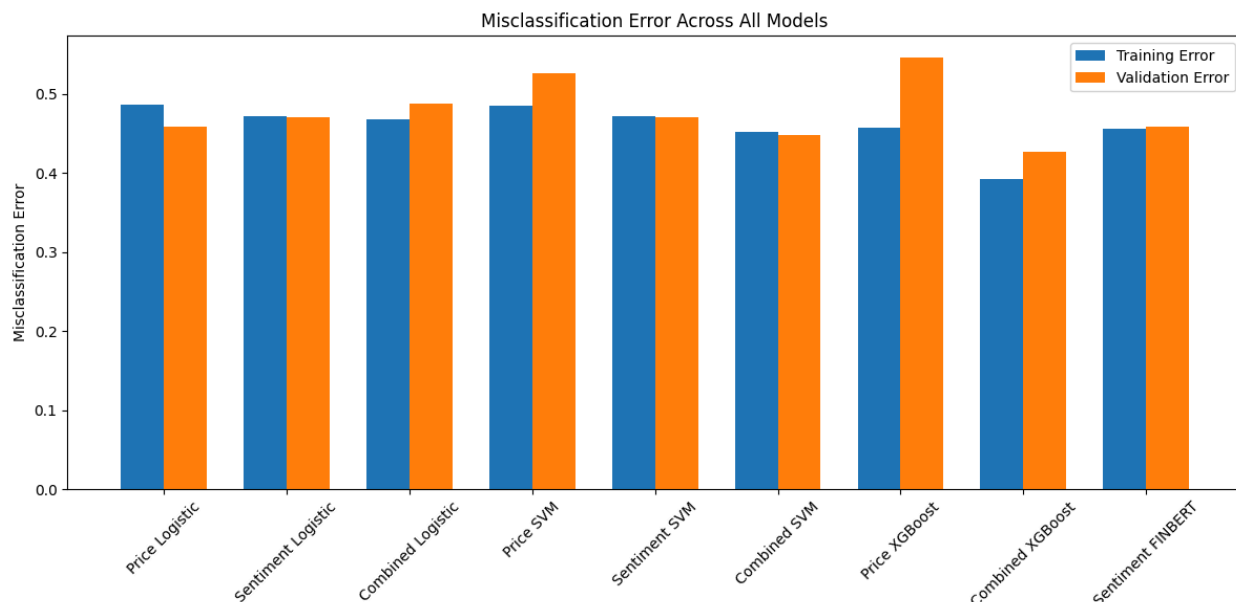
In previous weeks, we trained nine separate models and evaluated performance on a validation set. Three models (logistic, support vector machine, XGBoost) used solely historical AAPL stock price data, three models (logistic, support vector machine, FINBERT) used solely news sentiment data on the AAPL stock, and three models (logistic, support vector machine, XGBoost) used a combined dataset of both sentiment and historical price data. The training and validation set performance is below, in increasing order of model complexity:

	Training Misclassification Error (1-Accuracy)	Validation Misclassification Error (1 - Accuracy)	Training F1 (Macro Average)	Validation F1 (Macro Average)
Price Logistic	0.4856	0.4582	0.4301	0.4512
Sentiment Logistic	0.4717	0.4701	0.3457	0.3464
Combined Logistic	0.4678	0.4878	0.4740	0.4577
Price SVM	0.4848	0.5259	0.5152	0.4740
Sentiment SVM	0.4717	0.4702	0.3457	0.3464
Combined SVM	0.4521	0.4472	0.4409	0.4359
Price XGBoost	0.4569	0.5458	0.5367	0.4463
Combined XGBoost	0.3919	0.4268	0.5225	0.4590
Sentiment FINBERT	0.4552	0.4582	0.4174	0.4594

The primary metric used for best model consideration is accuracy with F1 macro average in support. Accuracy is the primary metric because a correct prediction in the increase or decrease of stock price is equally valuable. In the initial assumptions of our product, we assumed that the client made equal profit through investing (associated with an increase in the AAPL stock price) or divesting (associated with a decrease in the AAPL stock price). Therefore, we care most about the percent of correct overall predictions, followed by a metric that balances performance for each stock direction (f1 score). The misclassification error can also be viewed across models in the chart below.

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The XGBoost model on the combined dataset had the highest accuracy and lowest misclassification error on the validation dataset. It is not surprising that a model using the combined dataset had the highest accuracy, this model has more information available than any model using the sentiment or historical price datasets. In addition, the XGBoost model is the most complex model of the three using the combined dataset, and the sole model using decision tree methodology, which was improved even further with boosting. This model also likely succeeded because it was able to learn quickly on a small dataset of approximately two thousand observations, where other complex models (like FINBERT) took longer to understand the patterns between features.

In general, the more complex models tended to have lower errors on the validation datasets. In addition, the models using only historical price tended to perform the worst followed by sentiment data. This is likely due to the complex nature of the dataset needing complex models to understand patterns, and the sentiment and historical price datasets lacking critical information when used individually.

Model	Training Accuracy	Validation Accuracy	Testing Accuracy
Combined XGBoost	60.81%	57.32%	52.22%

This model decreased substantially in accuracy between the training, validation, and test datasets. This shows signals of high variance and is likely due to the model overfitting on the training dataset, which scaled poorly as the datasets changed.

We initially set a goal of 52% accuracy, which would achieve a \$71k annual return. This model surpassed that goal. This model is indeed good enough for the prediction task, but it could benefit from a larger training dataset to limit overfitting.